

Appointment

Feb 2024 - Present	Postdoctoral Fellow Supervisor: Prof. Roger Peng	University of Texas at Austin
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Education

Feb 2020 - Dec 2023	PhD in Statistics Thesis: <i>New tools for visualising and explaining multivariate spatio-temporal data</i> Supervisors: Prof. Dianne Cook, Dr. Patricia Menéndez, Dr. Nicolas Langrené, and Dr. Ursula Laa	Monash University, Australia
Feb 2016 - Dec 2019	Bachelor of Commerce with Honours Econometrics and Business Analytics, First Class Honours (H1) Thesis: <i>Exploration of judicial facial expression in videos of legal proceedings</i> Supervisors: Prof. Dianne Cook and Prof. Russell Smyth	Monash University, Australia

Awards

2026	IMS New Researcher Travel Award , <i>New Researcher Conference, Amherst, MA</i>
2025	KODAQS guest researcher visit program at LMU Munich Germany , 2,500 EUR
2025	Winner of the JSM Data Challenge Expo
2024	John M. Chambers Statistical Software Award <i>ASA Section of Statistical Computing and Statistical Graphics: 2,000 USD.</i> <i>Awarded for the R package, cubble, which introduces a new data structure for spatio-temporal data and provides ggplot2 extensions for creating glyph maps.</i>
2023	Posit Opportunity Scholar , <i>invitation to attend Posit::conf(2023) with all expenses covered</i>
2022	People's Choice Award , ECSS Miniconference, Statistical Society of Australia
2020 - 2023	Postgraduate (PhD) Research Scholarship , CSIRO Data61
2020 - 2023	Co-Funded Graduate Research Scholarship , Monash Business School
2020	Dean's Honours List , Monash Business School
2019	Summer Vacation Research Scholarship , Monash University

Publications

- Zhang, H. S.**, Cook, D., Langrené, N., & Leung, J. W. Y. (2026). Squintability and Other Metrics for Assessing Projection Pursuit Indexes and Guiding Optimization Choices. *Journal of Computational and Graphical Statistics*. <https://doi.org/10.1080/10618600.2026.2614076>
- Zhang, H. S.**, Peng, R. D. (2026). Inside Out: Externalizing Assumptions in Data Analysis as Validation Checks. *Journal of Data Science*. <https://doi.org/10.6339/25-JDS1206>
- Zhang, H. S.**, Cook, D., Laa, U., Langrené, N., & Menéndez, P. (2025). A Tidy Framework and Infrastructure to Systematically Assemble Spatio-temporal Indexes from Multivariate Data. *Journal of Computational and Graphical Statistics*, 1-19. <https://doi.org/10.1080/10618600.2024.2374960>
- Zhang, H. S.**, Cook, D., Laa, U., Langrené, N., & Menéndez, P. (2024). cubble: An R package for organizing and wrangling multivariate spatio-temporal data. *Journal of Statistical Software*, 110, 1-27. <https://doi.org/10.18637/jss.v110.i07>
- Lee, S., Cook, D., Silva, N. da, Laa, U., Spyrisson, N., Wang, E., & **Zhang, H. S.** (2022). The state-of-the-art on tours for dynamic visualization of high-dimensional data. *WIREs Computational Statistics*, 14(4), e1573. <https://doi.org/10.1002/wics.1573>

6. **Zhang, H. S.**, Cook, D., Laa, U., Langrené, N., & Menéndez, P. (2021). Visual Diagnostics for Constrained Optimisation with Application to Guided Tours. *The R Journal*, 13(2), 624–641. <https://doi.org/10.32614/RJ-2021-105>

Under revision:

7. Po, M., Yang, N., **Zhang, H. S.**, & Cook, D. (2026). Sugarglider: Create Glyph-Maps of Spatiotemporal Data. *The R Journal - accepted*. <https://github.com/maliny12/paper-sugarglider>
8. **Zhang, H. S.**, Reynolds, E., & Robbins, J. (2026). The ggplot2 extension ecosystem: an overview for users and developers. *Invited submission to WIREs Computational Statistics - minor revision*. <https://github.com/jtr13/extensions-review>
9. Coleman, S., **Zhang, H. S.**, Lucchesi, L., & Roy, S. (2026). Exploring the periodicity of flight patterns. *Invited submission to the CHANCE magazine - minor revision*.
10. **Zhang, H. S.**, Peng, R. D.(2025). An LLM-based Pipeline for Understanding Decision Choices in Data Analysis from Published Literature. <https://github.com/huizezhang-sherry/paper-decisions>

Software Activities

Package maintainer (primary author):

tidyindex	A general data pipeline for computing indexes
cubble	A data structure for organizing and wrangling spatio-temporal data
ferrn	Visual diagnostics plots for projection pursuit guided tour optimisation

Package contributor:

rj	Tools to help the associate editors with the reviewing process of the R journal
rjtools	Tools for authors to write, check, and submit articles to the R Journal
tourr	Tour methods for multivariate data visualisation

Other activities:

2024	maintainer of CRAN task view: Dynamic Visualizations
2024	reviewer of ROpenSci software peer review
2024	reviewer of John M. Chambers Statistical Software Award 2025
2025	contributor to fix bug in base R on array max.print

Teaching and Supervision

Instructor for UT Austin undergraduate coursework:

2025 Fall	SDS322E Elements of Data Sciences: https://huizezhang-sherry.github.io/SDS322E/
2026 Fall	SDS322E Elements of Data Sciences: https://huizezhang-sherry.github.io/SDS322E/

Research supervision:

2024	Google Summer of Code (GSoC) R Projects: Maliny Po: master student, now PhD student, Monash University Nathan Yang: undergraduate, Duke Univeristy
2024	Summer research project: Walter Wang: PhD student at UT Austin

TA for Monash undergraduate and graduate coursework:

2021, 2022	ETC5512 Wild Caught Data
2020, 2021	ETC5521 Exploratory Data Analysis
2019, 2020	ETC1010/ETC5510 Introduction to Data Analysis

Instructor for workshops:

2023 Apr	Switching between space and time: Spatio-temporal analysis with cubble, R Ladies Vienna
2022 Dec	Wrangling spatio-temporal data with R, WOMBAT link

Service

1. Associate editor for the R Journal, 2026 -
2. Program Chair (Elect), ASA Section of Statistical Graphics, 2026 (2025)
3. Journal review: R Journal, Journal of Data Science, The Australian & New Zealand Journal of Statistics, and Ecology and Evolution
4. Software review: ROpenSci and John M. Chambers Statistical Software Award review panel, 2024
5. Seminar organiser: Monash NUMBATs, 2022

Talks and Posters

2026, Jun	UseR! 2026	<i>Warsaw, Poland</i>
2026, Jun	LMU SODA Lab Group Seminar	<i>Munich, Germany</i>
2025 Jun	NYU-LMU-Workshop Making Data Work: Tools for Better Statistical Practice	<i>Munich, Germany</i>
2025 Aug	Joint Statistical Meeting (JSM)	<i>Nashville, USA</i>
2025 May	Symposium on Data Science and Statistics	<i>Salt Lake City, USA</i>
2024 Dec	University of Nebraska-Lincoln Graphic Group	<i>virtual</i>
2024 Oct	UT Austin SDS Department Seminar	<i>Austin, USA</i>
2024 Sep	Intl. Conf. on Trends & Perspectives in Linear Stats. Inference (Lin-Stat 2024)	<i>Poprad, Slovakia</i>
2024 Aug	Computational Statistics (COMPSTAT 2024)	<i>Giessen, Germany</i>
2024 Aug	Joint Statistical Meeting (JSM)	<i>Portland, USA</i>
2024 Aug	John M. Chambers Statistical Software Award Talk	<i>Portland, USA</i>
2024 Jan	Humboldt University of Berlin, Faculty of Economics	<i>Germany</i>
2023 Dec	Computational and Methodological Statistics (CMStatistics 2023)	<i>Berlin, Germany</i>
2023 Sep	Monash Research Tools Workshop	<i>Melbourne, Australia</i>
2023 May	University of Münster, Institute for Geoinformatics	<i>Germany</i>
2023 Feb	The Australian Spatial Econometrics and Statistics Workshop	<i>Melbourne, Australia</i>
2023 Apr	Maynooth University, Department of Mathematics and Statistics	<i>Ireland</i>
2022 Nov	CANSSI Ontario Statistical Software Conference	<i>virtual</i>
2022 Nov	Early Career and Student Statisticians Miniconference	<i>virtual</i>
2022 Jun	UseR! 2022	<i>virtual</i>
2021 Jul	UseR! 2021	<i>virtual</i>

Media Coverage

2023	Monash Graduate research impact: Crunching the numbers: How new research has revealed the hidden truth behind indexes
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Referees

1. **Roger D. Peng**, Postdoc Supervisor, roger.peng@austin.utexas.edu, Professor of Statistics, Department of Statistics and Data Sciences at the University of Texas at Austin, Texas, USA
2. **Dianne H. Cook**, PhD Thesis Advisor, dicook@monash.edu, Professor of Statistics, Department of Econometrics and Business Statistics at Monash University, Australia
3. **Nicolas Langrené**, PhD Thesis Advisor, nicolaslangrene@uic.edu.cn, Associate Professor in the UIC Financial Mathematics Programme at Beijing Normal-Hong Kong Baptist University, Zhuhai, Guangdong, China